

PERMUTATIONS AND COMBINATIONS

PERMUTATIONS

- Arrangement
- Order of the object is important
- The number of permutations of n things taken all at a time is $n!$.

$${}^n P_r = \frac{n!}{(n-r)!}$$

COMBINATIONS

- Selections
- Order in which objects are taken is not important

$${}^n C_r = \frac{n!}{r! (n-r)!}$$

Points to remember

- N things can be arranged in a circular arrangements in $(n-1)!$ ways.
- The number of straight lines that can be formed by joining n non-collinear points on the same plane is given by nC_2
- The number of triangles that can be formed by joining n non collinear points on the same plane is given by nC_3
- The number of handshakes is same as number of ways of selecting 2 people out of n which is equal to nC_2

Distinguish between Permutation and Combination

- In how many ways 4 persons can be seated in 6 chairs?
- In how many ways 4 persons can be selected from a group of 6 members?
- A person has 6 trousers and 9 shirts. In how many different ways he select a trouser and a shirt?
a) 15 b) 54 c) 6 d) 9
- How many four digit numbers can be formed using the digits (1,2,5,6,8,9)? (repetition of digits is not allowed)
a) 60 b) 24 c) 360 d) 120

- How many 4 digit odd numbers can be formed using the digits (2,4,6,8,9)? (repetition of digits is not allowed)

a)60

b)24

c)360

d)120

- In a party each person shook hands with every other person present. The total number of hand shakes was 36. Find the number of people present in the party?

a)18

b)9

c)8

d)10

- Find the number of ways of selecting 5 members from a group of 11 members so that one particular member is always included?

a) 210

b) 462

c) 420

d)330

- In how many ways can a cricket team of 11 members can be selected from 15 players,so that a particular player is included and another particular player is left out?

a) 216 b) 326 c) 426 d)286

- A committee of five members is to be selected from a group of 6 men and four women.In how many ways can the committee be formed so that it contains more men than women?

a)186 b)180 c)126 d)56